ON THE QUARTIC INVARIANT OF ODD DEGREE BINARY FORMS

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Abstract. We determine the squarefree part of the scalar factor that arises when the quartic invariant of the generic binary form F of odd degree $2n+1$ is expressed as the discriminant of the unique quadratic covariant $(F,F)_{2n}$. This squarefree part is exactly p when $n+2$ is a power of an odd prime p, and 1 otherwise. Equivalently, for each prime p: $v_2(S(n))$ is always even, and for odd p, $v_p(S(n))$ is odd if and only if $n+2$ is a power of p. This generalizes the classical identity $\mathrm{disc}(H(F)) = -3\cdot\mathrm{disc}(F)$ for binary cubics, which dates back to the work of Cayley and Sylvester in the 1850s. The proof, which involves substantial explicit coefficient analysis and p-adic deformation arguments, was developed using an AI-assisted research workflow: the author's earlier partial attempts were completed through systematic collaboration with Claude Code (Anthropic), and key arithmetic lemmas were formally verified in Lean 4 using Aristotle (Harmonic). We describe this workflow in detail as a case study in AI-assisted mathematical research. We also discuss representation-theoretic, geometric, and arithmetic interpretations of the quadratic covariant.

1. Introduction

A classical problem in invariant theory is to compute the scalar factors that arise when a given invariant or covariant of a binary form is constructed by two different methods. In this paper we determine the squarefree part of the scalar factor relating the quartic invariant of the generic odd-degree binary form to the discriminant of its unique quadratic covariant. The proof was completed using an AI-assisted workflow built around Claude Code (Anthropic) and Aristotle (Harmonic), and we describe this workflow in detail as a case study in AI-assisted mathematical research.

1.1. The classical case of binary cubics.

Let $F(x,z) = f_0x^3 + f_1x^2z + f_2xz^2 + f_3z^3 \in \mathbb{C}[x,z]$ be a binary cubic form. The *Hessian* of F is the binary quadratic form

$$H(F) := \frac{1}{4}\cdot\det\begin{pmatrix} \dfrac{\partial^2 F}{\partial x^2} & \dfrac{\partial^2 F}{\partial x\,\partial z} \\ \dfrac{\partial^2 F}{\partial z\,\partial x} & \dfrac{\partial^2 F}{\partial z^2}\end{pmatrix} = (3f_0f_2 - f_1^2)\,x^2 + (9f_0f_3 - f_1f_2)\,xz + (3f_1f_3 - f_2^2)\,z^2. \tag{1}$$

The form $H(F)$ is covariant under the action of $\mathrm{SL}_2(\mathbb{C})$ that sends $F(x,z) \mapsto F((x,z)\cdot\gamma)$ for each $\gamma \in \mathrm{SL}_2(\mathbb{C})$; moreover, it is the unique covariant of F with *degree* 2 (in the f_i) and *order* 2 (in x,z). The Hessian covariant was introduced by Cayley and studied extensively by Sylvester in the 1850s; we refer to [Ell95] and [GY03] for detailed classical treatments, and to [Olv99] for a modern account.

Since the discriminant of $H(F)$ is an $\mathrm{SL}_2(\mathbb{C})$-invariant of degree 4 and order 0, it must be a scalar multiple of $\mathrm{disc}(F)$. Indeed, a direct computation yields

$$\mathrm{disc}(H(F)) = -3\cdot\mathrm{disc}(F). \tag{2}$$

The specific scalar factor -3 for binary cubics appears in the classical texts of Elliott [Ell95] and Grace–Young [GY03]. More generally, a recurring theme in classical invariant theory is the determination of the precise scalar factors relating different constructions of the same covariant. For instance, McMahon [McM89] computed the scalar factor relating two expressions for the Hessian of a binary form of degree n.

The factor -3 appearing in (2) has deep number-theoretic significance. The Scholz reflection principle [Sch32] states that if $D<0$ is a fundamental discriminant, then the 3-rank of the class group of $\mathbb{Q}(\sqrt{D})$ is either equal to or one larger than that of $\mathbb{Q}(\sqrt{-3D})$. The identity (2) underlies the relationship between cubic fields and their quadratic resolvents: under the Delone–Faddeev correspondence, the Hessian maps a binary cubic of discriminant D to a quadratic form of discriminant $-3D$, and this mechanism is at the heart of the Davenport–Heilbronn density theorems for cubic fields [DH71] and their modern extensions [Bha04b, BST13, BV16].

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1.2. **The generalization to odd degree.** The objective of this paper is to determine the higher-degree analogue of the scalar factor -3 in (2) for binary forms of any given odd degree $2n+1$. The exact scalar is not itself a single integer but rather a product of factorial expressions; what we compute is its squarefree part, which turns out to be governed by the prime-power structure of $n+2$. Fix $n \geq 1$ and let

$$F(x,z) = \sum_{j=0}^{2n+1} f_j\, x^{2n+1-j} z^j \tag{3}$$

be the generic binary form of degree $2n+1$ with indeterminate coefficients $f_0, \ldots, f_{2n+1}$ over $\mathbb{Z}$. The *quadratic covariant* $Q_n = (F,F)_{2n}$ is the $(2n)$-th transvectant of F with itself (see §2 for the precise normalization). Transvectants, also known as *Überschiebungen* in the German literature, provide the fundamental algebraic mechanism for producing covariants from binary forms; we refer to Olver [Olv99, Chapter 5] and Kung–Rota [KR84] for comprehensive modern treatments, and to Elliott [Ell95] and Grace–Young [GY03] for the classical development. The form $Q_n = A_n x^2 + B_n xz + C_n z^2$ is a binary quadratic, unique up to scalar among degree-2, order-2 covariants of F (see §2.1).

The discriminant

$$\Delta_n := B_n^2 - 4A_nC_n \in \mathbb{Z}[f_0, \ldots, f_{2n+1}] \tag{4}$$

is the quartic invariant of F. We define the *content* $S(n) = \mathrm{cont}(\Delta_n)$ to be the greatest common divisor of all integer coefficients of Δ_n, and write $\mathrm{sqf}(m)$ for the squarefree part of a nonzero integer m.

1.3. **Main result.** For an integer $m \geq 1$, define

$$a(m) := \begin{cases} p, & \text{if } m = p^k \text{ for some odd prime } p \text{ and } k \geq 1, \\ 1, & \text{otherwise.} \end{cases} \tag{5}$$

Theorem 1.1. *For every integer $n \geq 1$ and every prime p, the parity of $v_p(S(n))$ is determined as follows:*

(i) *$v_2(S(n))$ is always even.*

(ii) *For every odd prime p, $v_p(S(n))$ is odd if and only if $n+2$ is a power of p.*

Equivalently,

$$\mathrm{sqf}(S(n)) = a(n+2),$$

where sqf *denotes the squarefree part.*

Computationally, the sequence $\mathrm{sqf}(S(n))$ for $n = 1, 2, 3, \ldots$ begins $3, 1, 5, 1, 7, 1, 3, 1, 11, 1, 13, \ldots$, which is the sequence A155457 in the OEIS. Theorem 1.1 identifies this sequence completely. The theorem is accompanied by a Lean 4 formalization (≈ 6000 lines) and a companion Jupyter notebook that computationally verifies every stated formula and lemma. All materials are available at

`https://github.com/ashvin-swaminathan/quartic-invariant`

and described further in §7.

In p-adic terms, part (ii) of Theorem 1.1 is equivalent to: for every odd prime p,

$$v_p(S(n)) \equiv \begin{cases} 1 \pmod 2, & \text{if } n+2 = p^k, \\ 0 \pmod 2, & \text{otherwise.} \end{cases} \tag{6}$$

Example 1.2. For small values of n:

- $n = 1$ ($m = 3$): $\mathrm{sqf}(S(1)) = 3$. This recovers the factor -3 in (2).
- $n = 2$ ($m = 4$): $\mathrm{sqf}(S(2)) = 1$, since $4 = 2^2$ is a power of the even prime 2.
- $n = 3$ ($m = 5$): $\mathrm{sqf}(S(3)) = 5$.
- $n = 4$ ($m = 6$): $\mathrm{sqf}(S(4)) = 1$, since 6 is not a prime power.

Remark. It is initially surprising that the prime-power criterion involves $n+2$ rather than n itself. This shift arises naturally from the binomial coefficients $\binom{n+2}{r}$ that appear when one simplifies the factorial expressions in the transvectant; see the proof of Lemma 3.3, where the substitution $m = n+2$ converts the factorial product into a squared factor times $E_m(r)$.

Remark. The reason $p = 2$ is excluded from part (ii) is structural: the factor of 4 in $\Delta_n = B_n^2 - 4A_nC_n$ ensures that even in the prime-power case $m = 2^k$, the 2-adic valuation picks up $+2$ rather than $+1$, keeping the parity even; see Proposition 5.10.

1.4. **Outline of the proof.** The proof of Theorem 1.1 occupies Sections 2–5, and proceeds as follows.

(1) In Section 2 we fix the transvectant normalization, derive explicit two-term formulas for the linear forms D_i and E_i appearing in the transvectant sum, and state the resulting formulas for A_n, B_n, C_n.
(2) In Section 3 we expand the discriminant $\Delta_n = B_n^2 - 4A_nC_n$ and compute, in closed form, the coefficient $C_{n,r}$ of a controlled family of test monomials $M_{n,r} = f_a f_{a+1} f_b f_{b+1}$. We then factor $C_{n,r}$ to isolate a non-square factor $E_m(r)$ and prove that $\gcd_r E_m(r)$ is a power of 2.
(3) In Section 4 we show that the content $S(n)$ divides the gcd of the $C_{n,r}$, prove the binomial gcd theorem $g(m) = \gcd_{1\le r\le m-1} \binom{m}{r}$ via Kummer's and Lucas' theorems, and show that the odd part of the gcd is governed by $g(m)$.
(4) In Section 5 we complete the proof of the prime-to-3 theorem by analyzing the p-adic parity of $v_p(S(n))$ for each odd prime $p \ge 5$. We show that primes dividing n contribute even valuation; for the non-prime-power case we use cancellation-free B_n coefficients; and for the prime-power case we establish a rank-1 collapse of $Q_n^{\#}$ modulo p followed by a first-order deformation argument. The same prime-power argument also treats the case $p = 3$ when $n + 2$ is a power of 3. In the same section, we show that $v_2(S(n))$ is always even; this follows from the fact that the discriminant formula $\Delta_n = B_n^2 - 4A_nC_n$ has a built-in factor of 4 that shifts the 2-adic deformation valuation by $+2$ instead of $+1$.

Section 7 describes the AI-assisted workflow used to complete this project.

1.5. **History and methodology.** This problem was suggested to the author by Manjul Bhargava as one of the first questions to consider at the start of the author's PhD. The author worked on it for several weeks and developed the basic framework that appears in this paper: the identification of the test monomial family $M_{n,r}$ and its connection to binomial coefficients, the role of Lucas' and Kummer's theorems, and the even-valuation phenomenon for primes dividing n. However, completing the proof required substantial technical work—particularly the cancellation-free coefficient analysis, the rank-1 collapse argument in the prime-power case, and the first-order deformation that forces odd p-adic valuation—and the project stalled.

The proof was completed using an AI-assisted workflow built around Claude Code (Anthropic's command-line AI tool). This workflow proceeded in several stages:

(i) *Numerical experimentation.* Claude Code wrote Python scripts that computed $S(n)$ for $n \le 30$, verified all coefficient formulas, and tested proof strategies by evaluating explicit specializations of the generic form. These experiments confirmed the theorem and identified the precise mechanism (the p-adic deformation at distance $t = p^{k-1}$) that forces odd valuation in the prime-power case.
(ii) *Proof construction.* Working from the author's earlier notes, Claude Code helped develop complete proofs for the missing cases, particularly the composite case involving $p = 3$ (where the end-anchored witness approach fails, but the cancellation-free B_n method works uniformly) and the general prime-power case $m = p^k$ for $k \ge 2$.
(iii) *Formal verification.* Key lemmas were submitted to Aristotle (Harmonic's Lean 4 formalization tool), which produced machine-checked proofs of the binomial GCD theorem, the $E_m(r)$ gcd lemma, the p^2-divisibility result, and the cancellation-free B_n coefficient formula.
(iv) *Exposition.* The present paper was largely written by Claude Code, with the author providing the original mathematical notes, directing the overall narrative, and reviewing the output. The introduction and interpretive sections draw on the author's framing of the problem; the technical exposition was generated by Claude Code from the proof constructions in the earlier phases.

The project is notable not for the depth of the final result—which is a concrete but somewhat technical computation in classical invariant theory—but for the fact that the workflow described above was able to handle it at all. The proof involves pages of intricate algebraic manipulation (tracking signs, factorial cancellations, binomial coefficient identities), careful p-adic analysis with case splits for different primes and prime powers, and explicit specialization arguments that must be tailored to the structure of the problem. This is precisely the kind of computation-heavy, detail-sensitive mathematical work that one might expect to be most resistant to AI assistance, and the success of the approach suggests that AI-assisted workflows may be effective for a broader class of research problems in number theory and algebra.

2. Preliminaries and normalization

We begin by recalling the definition and normalization of the transvectant operation, which is the central tool in our construction. The transvectant (German: *Überschiebung*) was developed by Cayley, Sylvester, and their contemporaries in the mid-nineteenth century as the primary method for producing new covariants from old ones. The operation was placed on a firm representation-theoretic footing by the symbolic method; see Kung–Rota [KR84, §4] for a modern account of this classical technique, and Olver [Olv99, Chapter 5] for a thorough textbook treatment.

Definition 2.1 (Transvectant)**.** For binary forms G of degree d and H of degree e, the r-th transvectant is

$$(G,H)_r = \sum_{i=0}^{r} (-1)^i \binom{r}{i} \frac{\partial^r G}{\partial x^{r-i}\,\partial z^i}\,\frac{\partial^r H}{\partial x^i\,\partial z^{r-i}}. \tag{7}$$

This is the standard normalization with no extra $1/(r!)$ or $1/\binom{d}{r}\binom{e}{r}$ factor. We refer to [Olv99] and [Ell95] for background on transvectants and classical invariant theory. We note that some authors (e.g., [KR84]) include additional normalizing factors; throughout this paper, we use (7) without modification.

Throughout, $F(x,z) = \sum_{j=0}^{2n+1} f_j\, x^{2n+1-j} z^j$ is the generic form of degree $N+1 := 2n+1$, so $N = 2n$. We set $m = n+2$.

Lemma 2.2. *$Q_n = (F,F)_{2n} = A_n x^2 + B_n xz + C_n z^2$, where*

$$Q_n = \sum_{i=0}^{N} (-1)^i \binom{N}{i} D_i\, E_i, \tag{8}$$

with the two-term linear forms

$$D_i = \alpha_i\, f_i\, x + \beta_i\, f_{i+1}\, z, \tag{9}$$

$$E_i = \beta_i\, f_{N-i}\, x + \alpha_i\, f_{N+1-i}\, z, \tag{10}$$

and the factorials $\alpha_i = (2n+1-i)!\,i!$, $\beta_i = (2n-i)!\,(i+1)!$.

In particular,

$$A_n = \sum_{i=0}^{N} (-1)^i \binom{N}{i} \alpha_i\beta_i\, f_i\, f_{N-i}, \tag{11}$$

$$B_n = \sum_{i=0}^{N} (-1)^i \binom{N}{i} \big(\alpha_i^2\, f_i\, f_{N+1-i} + \beta_i^2\, f_{i+1}\, f_{N-i}\big), \tag{12}$$

$$C_n = \sum_{i=0}^{N} (-1)^i \binom{N}{i} \alpha_i\beta_i\, f_{i+1}\, f_{N+1-i}. \tag{13}$$

Proof. **Computing D_i.** A typical monomial of F is $f_j\, x^{2n+1-j} z^j$. Taking $\frac{\partial^N}{\partial x^{N-i}\partial z^i}$:

$$\frac{\partial^N (f_j x^{2n+1-j} z^j)}{\partial x^{N-i}\partial z^i} = f_j \cdot \frac{(2n+1-j)!}{(2n+1-j-(N-i))!} \cdot \frac{j!}{(j-i)!} \cdot x^{2n+1-j-(N-i)} z^{j-i}.$$

For this to be nonzero we need $j \le i+1$ and $j \ge i$, so only $j = i$ and $j = i+1$ survive.

For $j = i$: the result is $\alpha_i f_i \cdot x$. For $j = i+1$: the result is $\beta_i f_{i+1} \cdot z$. Hence $D_i = \alpha_i f_i x + \beta_i f_{i+1} z$.

Computation of E_i. Similarly, $\frac{\partial^N F}{\partial x^i \partial z^{N-i}}$ is nonzero only for $j = N-i$ and $j = N+1-i$, giving $E_i = \beta_i f_{N-i} x + \alpha_i f_{N+1-i} z$.

Expanding $D_i E_i$.

$$\begin{aligned} D_i E_i &= (\alpha_i f_i x + \beta_i f_{i+1} z)(\beta_i f_{N-i} x + \alpha_i f_{N+1-i} z) \\ &= \alpha_i\beta_i\, f_i f_{N-i}\, x^2 + (\alpha_i^2\, f_i f_{N+1-i} + \beta_i^2\, f_{i+1} f_{N-i})\, xz + \alpha_i\beta_i\, f_{i+1} f_{N+1-i}\, z^2. \end{aligned}$$

Substituting into (8) and collecting the x^2, xz, z^2 coefficients gives (11)–(13). □

Definition 2.3. We write $\Delta_n = B_n^2 - 4A_nC_n$ for the discriminant of Q_n, and $S(n) = \mathrm{cont}(\Delta_n)$ for the gcd of all integer coefficients of Δ_n viewed as a polynomial in $f_0, \ldots, f_{2n+1}$.

2.1. **Representation-theoretic uniqueness.** Let V denote the standard 2-dimensional representation of SL_2 over a field of characteristic 0, and identify binary forms of degree d with $\mathrm{Sym}^d(V^\vee)$. The Clebsch–Gordan decomposition (see [FH91, Lecture 11]) gives an SL_2-equivariant splitting

$$\mathrm{Sym}^d(V^\vee) \otimes \mathrm{Sym}^d(V^\vee) \;\cong\; \bigoplus_{i=0}^{d} \mathrm{Sym}^{2d-2i}(V^\vee). \tag{14}$$

When $d = 2n+1$ is odd, the summand $\mathrm{Sym}^2(V^\vee)$ occurs precisely for $i = d-1 = 2n$. In particular, $\mathrm{Sym}^2(V^\vee)$ occurs with *multiplicity* 1 in (14), and therefore there exists, up to an overall scalar, a unique SL_2-equivariant bilinear map

$$\pi_{2n} : \mathrm{Sym}^{2n+1}(V^\vee) \times \mathrm{Sym}^{2n+1}(V^\vee) \longrightarrow \mathrm{Sym}^2(V^\vee).$$

The $(2n)$-th transvectant $(\cdot,\cdot)_{2n}$ is a concrete realization of π_{2n}, and our normalization in §2 fixes the scalar. Thus the existence and canonicity of $Q_n(f) = (f,f)_{2n}$ is forced by representation theory: it is the *unique* quadratic covariant of order 2 arising quadratically from an odd-degree form.

Having established the explicit shape of the quadratic covariant Q_n and defined the content $S(n)$, we turn in the next section to expanding the discriminant $\Delta_n = B_n^2 - 4A_nC_n$ and extracting the coefficients of a carefully chosen family of test monomials that will control the p-adic valuation of $S(n)$.

3. Discriminant expansion and explicit coefficient extraction

We now expand the discriminant $\Delta_n = B_n^2 - 4A_nC_n$ and extract the coefficients of a controlled family of test monomials. These coefficients, computed in closed form in Lemma 3.2, factor into a squared part and a non-square factor $E_m(r)$ whose gcd over r is a power of 2 (Lemma 3.4).

3.1. **Test monomials.** We introduce the following family of test monomials, whose coefficients in Δ_n will control the p-adic valuation of the content.

Definition 3.1 (Test monomials)**.** For $2 \le r \le m-1$ (equivalently $2 \le r \le n+1$), define

$$M_{n,r} = f_a\, f_{a+1}\, f_b\, f_{b+1}, \quad a = n+1-r, \quad b = n-1+r.$$

Note $a+b = 2n = N$, so $N-a = b$ and $N-b = a$. Also $a+1 < b$ when $r \ge 2$, so $M_{n,r}$ involves four distinct indeterminates. Its coefficient in Δ_n is denoted $C_{n,r}$.

3.2. **Closed formula for $C_{n,r}$.** The main computation of this subsection is the following closed-form expression for the coefficient of each test monomial.

Lemma 3.2. *For $2 \le r \le n+1$,*

$$C_{n,r} = -8\,(N!)^2\,(n+r)!\,(n+r-1)!\,(n-r+2)!\,(n-r+1)!\,\big(2n^2+4n+2r^2-4r+3\big). \tag{15}$$

This identity can be verified for specific values of n and r using the companion notebook.

Proof. The monomial $M_{n,r} = f_a f_{a+1} f_b f_{b+1}$ can appear in $\Delta_n = B_n^2 - 4A_nC_n$ from two sources.

Contribution from $-4A_nC_n$. In A_n, the monomial f_af_b arises from indices $i=a$ and $i=b$, giving $[f_af_b]\,A_n = (-1)^a\binom{N}{a}\alpha_a\beta_a + (-1)^b\binom{N}{b}\alpha_b\beta_b$. Similarly $[f_{a+1}f_{b+1}]\,C_n$ equals the same expression. Hence

$$[M_{n,r}](-4A_nC_n) = -4K^2, \quad K = (-1)^a\binom{N}{a}\alpha_a\beta_a + (-1)^b\binom{N}{b}\alpha_b\beta_b. \tag{16}$$

Since $b-a = 2r-2$ is even, $(-1)^b = (-1)^a$ and $\binom{N}{b} = \binom{N}{a}$. The factorials satisfy $\alpha_a\beta_a = (n+r)!(n+1-r)!(n+r-1)!(n+2-r)! = \alpha_b\beta_b$. A direct computation gives

$$\binom{N}{a}\alpha_a\beta_a = N!\,(n+r)!\,(n+2-r)!, \tag{17}$$

and therefore $K = -2(-1)^{n-r}\,N!\,(n+r)!\,(n-r+2)!$, so

$$4K^2 = 16\,(N!)^2\,(n+r)!^2\,(n-r+2)!^2. \tag{18}$$

Contribution from B_n^2. We have $[M_{n,r}]\,B_n^2 = 2UV$, where $U = [f_af_{b+1}]\,B_n$ and $V = [f_{a+1}f_b]\,B_n$.

Using the notation $P_i = (-1)^i \binom{N}{i} \alpha_i^2$ and $R_i = (-1)^i \binom{N}{i} \beta_i^2$, which simplify to

$$P_i = (-1)^i\, N!\,(N-i)!\,i!\,(N+1-i)^2,$$
$$R_i = (-1)^i\, N!\,(N-i)!\,i!\,(i+1)^2,$$

one finds that U receives four terms ($T_1 = P_a$, $T_2 = P_{b+1}$, $T_3 = R_{a-1}$, $T_4 = R_b$), which combine to give

$$U = -2(-1)^{n-r}\,(2r-1)\,N!\,(n+r)!\,(n-r+1)!. \tag{19}$$

Similarly, V receives four terms ($S_1 = P_{a+1}$, $S_2 = P_b$, $S_3 = R_a$, $S_4 = R_{b-1}$), giving

$$V = 2(-1)^{n-r}(2r-3)\,N!\,(n+r-1)!\,(n-r+2)!. \tag{20}$$

Combining. Set $P_0 = (N!)^2(n+r)!(n+r-1)!(n-r+2)!(n-r+1)!$. Then $2UV = -8(2r-1)(2r-3)\,P_0$ and $4K^2 = 16(n+r)(n-r+2)\,P_0$. Expanding:

$$\begin{aligned} &-8(4r^2-8r+3) - 16(n^2+2n-r^2+2r) \\ &= -8(2n^2+4n+2r^2-4r+3). \end{aligned}$$

This yields (15). □

3.3. **Factoring $C_{n,r}$.** We now factor the closed-form expression for $C_{n,r}$ to isolate the arithmetic content.

Lemma 3.3. *With $m = n+2$, define*

$$E_m(r) = (m-r)(m+r-2)\big(2(m-1)^2 + 2(r-1)^2 - 1\big). \tag{21}$$

Then

$$C_{n,r} = -8\big[N!\,(m+r-3)!\,(m-r-1)!\big]^2 \cdot E_m(r). \tag{22}$$

In particular, for any odd prime p,

$$v_p(C_{n,r}) = 2\,v_p\big(N!(m+r-3)!(m-r-1)!\big) + v_p(E_m(r)), \tag{23}$$

so $v_p(C_{n,r}) \equiv v_p(E_m(r)) \pmod 2$.

Proof. Substituting $n = m-2$ into the quadratic factor gives $2n^2+4n+2r^2-4r+3 = 2(m-1)^2+2(r-1)^2-1$. The factorial product factors as $(n+r)!(n+r-1)!(n-r+2)!(n-r+1)! = (m+r-2)(m-r)\cdot[(m+r-3)!(m-r-1)!]^2$, which yields (22). □

Lemma 3.4. *For any odd prime p and $m \geq 7$, $p \nmid \gcd_{2\leq r\leq m-1} E_m(r)$.*

Proof. For $p \mid E_m(r)$, at least one of the three factors must vanish modulo p: (i) $r \equiv m$, (ii) $r \equiv 2-m$, or (iii) $2(m-1)^2 + 2(r-1)^2 \equiv 1 \pmod p$.

For $p \geq 5$: conditions (i) and (ii) each exclude one residue class, and (iii) excludes at most two, for a total of at most 4 excluded classes out of $p \geq 5$. So some r avoids all three.

For $p = 3$: a case-by-case analysis on m mod 3 shows that at least one residue class of r modulo 3 avoids all conditions. Specifically:

- $m \equiv 0$: the class $r \equiv 1$ works.
- $m \equiv 1$: the classes $r \equiv 0$ and $r \equiv 2$ work.
- $m \equiv 2$: the class $r \equiv 1$ works.

For $m \geq 7$, the interval $\{2, \ldots, m-1\}$ has length at least 5, so it contains representatives of all three residue classes modulo 3. □

Having established the closed-form expression for the test coefficients $C_{n,r}$ and isolated the non-square factor $E_m(r)$, we now relate the gcd of the $C_{n,r}$ to the content $S(n)$ through the classical theory of binomial coefficient divisibility.

4. From explicit coefficients to the content

Since $S(n)$ divides every coefficient of Δ_n, we have $S(n) \mid \gcd_r C_{n,r}$. By (23) and Lemma 3.4, for each odd prime p the minimum of $v_p(C_{n,r})$ over r is even, so the odd part of the gcd is governed by the binomial coefficients embedded in the factorial structure. We make this precise via the classical binomial gcd theorem, which identifies $g(m) = \gcd_{1\leq r\leq m-1} \binom{m}{r}$ as p when m is a prime power p^k and 1 otherwise.

4.1. **The binomial gcd theorem.** The key input from elementary number theory is the following classical result.

Lemma 4.1 (Binomial gcd)**.** *For $m \geq 2$,*

$$g(m) := \gcd_{1 \leq r \leq m-1} \binom{m}{r} = \begin{cases} p, & \text{if } m = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 1, & \text{otherwise.} \end{cases}$$

Remark. This is a well-known result, sometimes called the "prime-power characterization of the gcd of binomial coefficients." It follows from a combination of two classical results in number theory: Kummer's theorem [Kum52] on the p-adic valuation of binomial coefficients, and Lucas' theorem [Luc78] on binomial coefficients modulo a prime. We include the proof for completeness and because we need the precise statement for the content analysis that follows.

Proof. **Case $m = p^k$.** By Kummer's theorem [Kum52], $v_p\binom{p^k}{r}$ equals the number of carries when adding r and $p^k - r$ in base p. For any $0 < r < p^k$, at least one carry occurs (at the lowest nonzero digit position of r), so $p \mid \binom{p^k}{r}$. For $r = p^{k-1}$, exactly one carry occurs, so $v_p\binom{p^k}{p^{k-1}} = 1$. Hence $g(p^k) = p$.

Case m not a prime power. For each prime $p \mid m$, since m is not a p-power, m has at least two nonzero base-p digits. Let $m_j > 0$ at a position j that is not the leading position. By Lucas' theorem [Luc78], taking $r = p^j$ gives $\binom{m}{p^j} \equiv m_j \pmod{p}$, with $0 < m_j < p$, so $p \nmid \binom{m}{r}$. For primes $p \nmid m$, m is not a p-power, so the same argument applies (the units digit of m in base p is nonzero). Hence $g(m) = 1$. □

4.2. **Relating the gcd to $S(n)$.** Combining the results of the previous subsection with the binomial gcd, we obtain the following.

Remark. By (23), $v_p(C_{n,r}) \equiv v_p(E_m(r)) \pmod{2}$ for each r, and Lemma 3.4 shows that $E_m(r)$ does not contribute any *uniform* odd prime factor to the gcd of the $C_{n,r}$. However, this does not by itself determine the parity of $v_p(\gcd_r C_{n,r})$, since a different r could in principle achieve a smaller valuation of opposite parity. The definitive parity analysis is carried out in §5.2 and §5.3 using explicit coefficient witnesses.

With the binomial gcd theorem and the parity observations in Remark 4.2 as motivation, we now turn to the heart of the proof: the p-adic parity analysis that distinguishes the prime-power case from the non-prime-power case.

5. p-ADIC PARITY AND PROOF OF THE MAIN THEOREM

We now analyze the p-adic valuation of $S(n)$ for each prime p, treating three cases according to the relationship between p and $m = n + 2$: primes dividing n (§5.1), the non-prime-power case (§5.2), and the prime-power case (§5.3). We then assemble the proof of Theorem 1.1 in §5.4, treat the prime 2 in §5.5, and record the prime-power case for $p = 3$ as well.

5.1. **Primes dividing n contribute even valuation.** We first handle the case where the prime divides n rather than m.

Lemma 5.1. *If p is an odd prime with $p \mid n$, then every coefficient of Q_n is divisible by p^2.*

Proof. We show $p \mid \alpha_i$ and $p \mid \beta_i$ for every $0 \leq i \leq N$. Since $p \mid n$ and p is odd, $p \leq n$. For $\alpha_i = (2n+1-i)!\, i!$: if $i \geq p$ then $p \mid i!$; if $i < p$ then $2n + 1 - i \geq 2n + 2 - p \geq n + 2 > p$, so $p \mid (2n + 1 - i)!$. Similarly for $\beta_i = (2n - i)!\,(i + 1)!$.

Since $p \mid \alpha_i$ and $p \mid \beta_i$, each $D_i \equiv 0$ and $E_i \equiv 0$ modulo p, giving $D_i E_i \equiv 0 \pmod{p^2}$. □

Remark. When $p \mid n$, Lemma 5.1 provides the stronger structural information that $e_p \geq 2$ (i.e., every coefficient of Q_n is divisible by p^2, not just by p^{e_p} for some $e_p \geq 1$). The even-parity conclusion $v_p(S(n)) = 2e_p$ follows from Proposition 5.3 below, which handles all non-prime-power cases uniformly.

Remark. For odd p: $p \mid n$ implies $p \nmid m = n + 2$, so the conditions "$p \mid n$" and "$p \mid m$" are disjoint.

5.2. **The non-prime-power case.** We now show that $v_p(S(n))$ is even whenever m is not a power of p.

Remark. One might hope to show $p \nmid S(n)$ when $p \nmid n(n+2)$, but this is false in general: for instance, $v_3(S(2)) = 4$ even though $3 \nmid 2 \cdot 4$. The correct statement is that $v_p(S(n))$ is *even* for all primes p with m not a p-power, which is proved below using the cancellation-free b_k witness.

Lemma 5.2. *For $1 \le k \le N$,*

$$b_k := [f_k f_{N+1-k}]\, B_n = 2(-1)^k\,(N!)^2\,\frac{(N+1-k)(N+1-2k)}{\binom{N}{k}}. \tag{24}$$

This identity can be verified for specific values of n using the companion notebook.

Proof. The monomial $f_k f_{N+1-k}$ arises from four summands in (12). All four share the same squared factorial α_k^2 (using $\beta_{k-1} = \alpha_k$, $\alpha_{N+1-k} = \alpha_k$, $\beta_{N-k} = \alpha_k$). Since $N = 2n$ is even: $(-1)^{N+1-k} = -(-1)^k$ and $(-1)^{N-k} = (-1)^k$. Also $\binom{N}{N+1-k} = \binom{N}{k-1}$. The four terms sum to

$$b_k = 2(-1)^k \alpha_k^2 \Big[\binom{N}{k} - \binom{N}{k-1}\Big] = 2(-1)^k \alpha_k^2 \binom{N}{k} \cdot \frac{N+1-2k}{N+1-k}.$$

Simplifying $\alpha_k^2 \binom{N}{k}/(N+1-k) = (N+1-k)(N-k)!\,k!\,N! = (N+1-k)(N!)^2/\binom{N}{k}$ gives (24). □

Proposition 5.3. *If $p \ge 5$ is an odd prime and $m = n+2$ is not a power of p, then $v_p(S(n))$ is even.*

Proof. We give a uniform argument that works regardless of whether $p \mid n$, $p \mid m$, or neither. Recall from (24) the cancellation-free coefficient $b_k = 2(-1)^k (N!)^2 (N+1-k)(N+1-2k)/\binom{N}{k}$. Since $N = 2n$ is even, $N+1$ is odd, so $N+1-2k \neq 0$ for every integer k; likewise $N+1-k \neq 0$ for $k \le N$. Hence $b_k \neq 0$ for all $1 \le k \le N$.

Define $e_p = \min_{1 \le k \le N} v_p(b_k)$. Every scalar coefficient of A_n, B_n, C_n has p-adic valuation at least e_p (the b_k are scalar coefficients of B_n, and a comparison of the $\alpha\beta$-scalars in A_n, C_n and the α^2/β^2-scalars in B_n shows they are all at least as large), so $\Delta_n = p^{2e_p} \Delta_n^{\#}$.

We exhibit a coefficient of Δ_n with p-adic valuation exactly $2e_p$. It suffices to find $k_0 \in \{1, \dots, N\} \setminus \{n, n+1\}$ with $v_p(b_{k_0}) = e_p$, because then the square monomial $(f_{k_0} f_{N+1-k_0})^2$ appears in Δ_n with coefficient $b_{k_0}^2$ and valuation $2e_p$ (see below for why $A_n C_n$ does not contribute).

Existence of k_0 for $p \ge 5$. Using (**??**) and the identity $\binom{N}{n-1} = \binom{N}{n} \cdot n/(n+1)$, one computes

$$v_p(b_{n+j}) - v_p(b_n) = v_p\Big(\textstyle\prod_{i=1}^{j}(n-i+1)\Big) - v_p\Big(\textstyle\prod_{i=1}^{j}(n+i+1)\Big) + v_p(N+1-(n+j)) + v_p(N+1-2(n+j)) - v_p(n+1) \tag{25}$$

for $j \ge 1$ (and similarly for negative shifts). For $j = -1$ (i.e., $k = n-1$), the formula simplifies to:

$$v_p(b_{n-1}) - v_p(b_n) = v_p(m) - v_p(n) + v_p(3).$$

For $p \ge 5$: $v_p(3) = 0$, and since p is odd and $m = n+2$, at most one of n and m is divisible by p.

Case $p \nmid m$: $v_p(m) = 0$, so $v_p(b_{n-1}) - v_p(b_n) = -v_p(n) \le 0$. Thus $v_p(b_{n-1}) \le v_p(b_n)$. If the minimum e_p is attained at $k = n$ or $k = n+1$, then this inequality shows it is also attained at the off-center index $k = n-1$. Otherwise the minimum is already attained at some off-center index. In either case there exists $k_0 \notin \{n, n+1\}$ with $v_p(b_{k_0}) = e_p$.

Case $p \mid m$: Write $a = v_p(m) \ge 1$ and $m = p^a r$ with $p \nmid r$ and $r \ge 2$ (since m is not a p-power). Set $k_0 = n - p^a$. Since $p^a \le m/2 = (n+2)/2 < n$ (using $r \ge 2$), we have $1 \le k_0 < n$, so $k_0 \in \{1, \dots, N\} \setminus \{n, n+1\}$.

We claim $v_p(b_{k_0}) \le v_p(b_n)$. From the closed form (24), the ratio is

$$\frac{b_{k_0}}{b_n} = \frac{(n+1+p^a)(2p^a+1)}{(n+1)\cdot 1} \cdot \frac{\binom{N}{n}}{\binom{N}{n-p^a}}.$$

Since $n+1 = p^a r - 1 \equiv -1$ and $n+1+p^a = p^a(r+1) - 1 \equiv -1 \pmod{p}$, and $2p^a + 1 \equiv 1 \pmod{p}$, the linear factors contribute $v_p = 0$. For the binomial ratio, write

$$\frac{\binom{N}{n}}{\binom{N}{n-p^a}} = \prod_{j=0}^{p^a-1} \frac{n+j+1}{n-j}.$$

Both the numerator set $\{n+1,\dots,n+p^a\}$ and the denominator set $\{n-p^a+1,\dots,n\}$ form complete residue systems modulo p^a. The unique multiple of p^a in the numerator is $p^a r$, with $v_p = a + v_p(r) = a$. The unique multiple of p^a in the denominator is $p^a(r-1)$, with $v_p = a + v_p(r-1)$. All other terms in both sets are non-multiples of p^a and contribute equal p-adic valuation (since they share the same residues modulo p^a). Hence

$$v_p\left(\frac{\binom{N}{n}}{\binom{N}{n-p^a}}\right) = v_p(r) - v_p(r-1) = -v_p(r-1) \le 0,$$

giving $v_p(b_{k_0}) \le v_p(b_n)$. Thus e_p is achieved at the off-center index k_0.

Now consider the square monomial $(f_{k_0} f_{N+1-k_0})^2$ in $\Delta_n = B_n^2 - 4A_nC_n$. Since $k_0 \ne n$ and $k_0 \ne n+1$, we have $k_0 \ne N-k_0$ and $k_0 \ne N+1-k_0$ (as $N=2n$), so the quartic monomial $f_{k_0}^2 f_{N+1-k_0}^2$ cannot arise from A_nC_n: every term of A_nC_n is a product $f_i f_{N-i} f_{j+1} f_{N+1-j}$ with $i \ne N-i$ (for $i \ne n$) and $j+1 \ne N+1-j$ (for $j \ne n$), so matching $f_{k_0}^2$ requires $i = N-i = k_0$, i.e., $k_0 = n$. Hence

$$[(f_{k_0} f_{N+1-k_0})^2]\,\Delta_n = b_{k_0}^2, \qquad v_p(b_{k_0}^2) = 2e_p.$$

Since every coefficient of Δ_n has $v_p \ge 2e_p$, and this particular coefficient has $v_p = 2e_p$, we conclude $v_p(S(n)) = 2e_p$, which is even. □

5.3. **The prime-power case.** Let $m = p^k$, $n = p^k - 2$, $N = 2p^k - 4$. Define $e_p = \min\{v_p(c) : c \text{ a scalar coefficient of } Q_n\}$. Then $Q_n = p^{e_p} Q_n^{\#}$ and $\Delta_n = p^{2e_p}\Delta_n^{\#}$.

Lemma 5.4. *The p-adic content e_p is attained by exactly the following summands of Q_n:*

(i) *the A_n/C_n summand at $i=n$, with scalar $\binom{N}{n}\alpha_n\beta_n$;*
(ii) *the β^2-part of the B_n summand at $i=n-1$, with scalar $\binom{N}{n-1}\beta_{n-1}^2$;*
(iii) *both parts of the B_n summand at $i=n$;*
(iv) *the α^2-part of the B_n summand at $i=n+1$, with scalar $\binom{N}{n+1}\alpha_{n+1}^2$.*

Every other scalar coefficient of A_n, B_n, or C_n has p-adic valuation at least e_p+1.

Proof. Since $\alpha_n = \beta_n = (n+1)!\,n! =: \gamma$, the summand at $i=n$ has valuation $v_p(\binom{N}{n}\gamma^2)$ in all three families. We check that this is the global minimum.

For the $\alpha\beta$-family (governing A_n and C_n): the scalar is $\binom{N}{i}\alpha_i\beta_i = N!\,(N+1-i)!\,(i+1)!$. At $i=n$, both auxiliary factorials equal $(n+1)!$. For $|i-n| \ge 1$, one of $N+1-i$ or $i+1$ exceeds $n+1 = p^k-1$, so either $(N+1-i)!$ or $(i+1)!$ contains a factor of $m = p^k$, increasing the valuation. (Precisely: if $i \le n-1$ then $N+1-i \ge n+2 = m$, and if $i \ge n+1$ then $i+1 \ge n+2 = m$.) Thus the $\alpha\beta$-minimum is attained only at $i=n$.

For the α^2-family (one part of B_n): the scalar is $\binom{N}{i}\alpha_i^2 = N!\,(N+1-i)^2(N-i)!\,i!$. At $i=n$ and $i=n+1$, one verifies $v_p = e_p$; for $|i-n| \ge 2$ or $i \le n-1$ the extra factor of m in $(N+1-i)!$ or $i!$ forces $v_p \ge e_p+1$.

For the β^2-family: the scalar is $\binom{N}{i}\beta_i^2 = N!\,(i+1)^2(N-i)!\,i!$. By the same analysis with i and $N-i$ exchanged, the minimum is attained at $i=n$ and $i=n-1$.

Combining: the global minimum e_p is achieved precisely at the four summand types listed in the statement, and all others have valuation at least e_p+1. □

Lemma 5.5. $Q_n^{\#} \equiv \lambda\,(f_n x - f_{n+1} z)^2 \pmod p$ *with $p \nmid \lambda$. More precisely, $Q_n^{\#}$ has rank 1 modulo p, so its discriminant vanishes: $p \mid \operatorname{cont}(\Delta_n^{\#})$.*

Remark. The fact that $Q_n^{\sharp}$ has rank 1 modulo p can be understood representation-theoretically: after reducing modulo p, only the central block of the transvectant sum (indices $i \in \{n-1, n, n+1\}$) survives by Lemma 5.4, and this block is forced to be a perfect square by the equality $\alpha_n = \beta_n$ that holds at the central index. See [FH91] for background on the representation theory of SL_2.

Proof. By Lemma 5.4, only $i \in \{n-1, n, n+1\}$ contribute modulo p. Set $\gamma = (n+1)!n!$.

At $i=n$: $\alpha_n = \beta_n = \gamma$, so the summand contributes $(-1)^n\binom{N}{n}\gamma^2(f_n x + f_{n+1} z)^2$.

At $i = n \pm 1$: since $\alpha_{n-1} = (n+2)!(n-1)!$ has extra factor p^k from $(n+2)!$, while $\beta_{n-1} = \gamma$, the α-terms vanish modulo $p \cdot p^{e_p}$. So $D_{n-1} \equiv \gamma f_n z$ and $E_{n-1} \equiv \gamma f_{n+1} x$; similarly $D_{n+1} \equiv \gamma f_{n+1} x$ and $E_{n+1} \equiv \gamma f_n z$. These contribute only to the xz coefficient.

Setting $\lambda_A = (-1)^n\binom{N}{n}\gamma^2/p^{e_p}$ (with $p \nmid \lambda_A$), we obtain $A_n^\# \equiv \lambda_A f_n^2$, $C_n^\# \equiv \lambda_A f_{n+1}^2$, and $B_n^\# \equiv \frac{2\lambda_A}{n+1} f_n f_{n+1} \equiv -2\lambda_A f_n f_{n+1} \pmod p$, where the last step uses $n+1 = p^k - 1 \equiv -1 \pmod p$. Thus $Q_n^\# \equiv \lambda_A(f_n x - f_{n+1} z)^2 \pmod p$.

The discriminant satisfies

$$(B_n^\#)^2 - 4A_n^\# C_n^\# \equiv 4\lambda_A^2 f_n^2 f_{n+1}^2 - 4\lambda_A^2 f_n^2 f_{n+1}^2 = 0 \pmod p.$$

Hence $Q_n^\#$ has rank 1 modulo p. □

Corollary 5.6. $p \mid \mathrm{cont}(\Delta_n^\#)$. *In particular,* $v_p(S(n)) \geq 2e_p + 1$.

First-order deformation. Set $t = p^{k-1}$. For $k \geq 2$, consider the specialization

$$f_n = 1, \quad f_{n+1} = -1, \quad f_{n-t} = s, \quad f_{n+t+1} = 1, \quad \text{others } 0. \tag{26}$$

For $k = 1$ ($t = 1$, $m = p$, $n = p - 2$):

$$f_{n-1} = s, \quad f_n = 1, \quad f_{n+1} = -1, \quad f_{n+2} = 1, \quad \text{others } 0. \tag{27}$$

Lemma 5.7. *For* $k \geq 2$, $v_p(\binom{N}{n-t}\alpha_{n-t}^2) = e_p + 1$.

Proof. The B_n coefficient b_{n-t} has $v_p(n+t+1) = v_p(p^k + p^{k-1} - 1) = 0$ and $v_p(2t+1) = v_p(2p^{k-1}+1) = 0$ (for $k \geq 2$).

A carry analysis via Kummer's theorem shows that $v_p\binom{N}{n-t} = v_p\binom{N}{n} - 1$: the carry chain breaks at position $k-1$ when adding $(n-t) + (n+t) = N$, losing one carry compared to $n + n = N$. Hence $v_p(b_{n-t}) = 2v_p(N!) - (c_p - 1) = e_p + 1$. □

Lemma 5.8. *For* $k \geq 2$, *or for* $k = 1$ *with* $p \geq 5$*: under the specialization,* $B^\# \equiv 2u + p\mu s \pmod{p^2}$ *and* $A^\# \equiv C^\# \equiv u \pmod p$, *where* $p \nmid u\mu$.

Proof. **Case $k \geq 2$.** At $s = 0$: with $f_n = 1$ and $f_{n+1} = -1$, Lemma 5.5 gives $Q_n^\# \equiv \lambda(x+z)^2 \pmod p$, so $u = \lambda$ and $B^\# \equiv 2u \pmod p$. The parameter $s = f_{n-t}$ enters through indices $i \in \{n-t, n+t, n+t+1\}$. All three contribute only to B_n (the xz coefficient), at valuation $e_p + 1$. By Lemma 5.7, the sum divided by p^{e_p+1} is a p-adic unit μ.

Case $k = 1$, $p \geq 5$. Here $t = 1$, and b_{n-1} has $v_p(N + 1 - 2(n-1)) = v_p(3) = 0$, so $v_p(b_{n-1}) = e_p + 1$ and $\mu \neq 0 \bmod p$.

Case $k = 1$, $p = 3$. This case is excluded from the present lemma and handled separately below as a base case for Proposition 5.9. □

Remark (Base case $m = 3$). When $p = 3$ and $k = 1$, we have $n = 1$ and $Q_1 = -2H(F)$, where $H(F)$ is the classical Hessian of a binary cubic. The identity (2) gives $\Delta_1 = 4\,\mathrm{disc}(H(F)) = -12\,\mathrm{disc}(F)$. Since $\mathrm{disc}(F)$ is a primitive polynomial in the f_j (its content is 1), and $v_3(12) = 1$, it follows that $S(1) = \mathrm{cont}(\Delta_1) = 12 \cdot \mathrm{cont}(\mathrm{disc}(F)) \cdot (\text{power of } 2)$. Direct computation gives $S(1) = 192 = 2^6 \cdot 3$, so $v_3(S(1)) = 1$, which is odd. This confirms Proposition 5.9 in this case.

Proposition 5.9. *When* $m = p^k$ *(p odd, $k \geq 1$),* $v_p(\mathrm{cont}(\Delta_n^\#)) = 1$.

Proof. For $k \geq 2$, or $k = 1$ with $p \geq 5$: from Lemma 5.8,

$$\begin{aligned}(B^\#)^2 - 4A^\# C^\# &\equiv (2u + p\mu s)^2 - 4u^2 = 4u^2 + 4up\mu s + p^2\mu^2 s^2 - 4u^2 \\ &\equiv 4up\mu s \pmod{p^2}.\end{aligned}$$

At $s = 1$: $v_p = 1$ (since $p \nmid u\mu$). Combined with Corollary 5.6: $v_p(\mathrm{cont}(\Delta_n^\#)) = 1$.

For $p = 3$, $k = 1$: by Remark 5.3, $v_3(S(1)) = 1$, so $v_3(\mathrm{cont}(\Delta_1^\#)) = 1$ as well. □

5.4. **Synthesis: proof of Theorem 1.1.** We now combine the results of the preceding subsections to establish the main theorem.

Proof of Theorem 1.1. **Part (i):** $v_2(S(n))$ is always even, by Proposition 5.10.

Part (ii): Fix an odd prime p.

If $m = p^k$: By Proposition 5.9, $v_p(S(n)) = 2e_p + 1 \equiv 1 \pmod 2$, so $p \mid \mathrm{sqf}(S(n))$. Since m can be a power of at most one odd prime, the odd part of $\mathrm{sqf}(S(n))$ is p.

If m is not a power of p: By Proposition 5.3 (for $p \geq 5$, and by Proposition 6.8 for $p = 3$), $v_p(S(n)) \equiv 0 \pmod 2$, so $p \nmid \mathrm{sqf}(S(n))$.

Combining parts (i) and (ii): $\mathrm{sqf}(S(n)) = a(n+2)$. □

Remark. The proof of part (ii) for $p = 3$ in the non-prime-power case uses a different witness construction from the $p \geq 5$ case, based on ternary digit recursions. This is developed in Section 6 below.

5.5. **The prime $p = 2$.** We show that $v_2(S(n))$ is always even, so the prime 2 never contributes to the squarefree part of $S(n)$.

Proposition 5.10. *For every $n \geq 1$, $v_2(S(n))$ is even.*

Proof. Case m not a power of 2. The cancellation-free argument of Proposition 5.3 applies verbatim: the valuation shift formula $v_2(b_{n-1}) - v_2(b_n) = v_2(m) - v_2(n) + v_2(3) = v_2(m) - v_2(n)$ (since $v_2(3) = 0$) gives an off-centre witness exactly as in §5.2 (for the case $2 \nmid m$) or the complete-residue-system argument (for $2 \mid m$, m not a 2-power). Hence $v_2(S(n)) = 2e_2$ is even.

Case $m = 2^k$ ($k \geq 2$; note $k = 1$ gives $n = 0$). The rank-1 collapse (Lemma 5.5) shows $Q_n^{\#} \equiv \lambda(f_n x - f_{n+1} z)^2 \pmod 2$. In particular $B^{\#} \equiv -2\lambda f_n f_{n+1} \equiv 0 \pmod 2$, so $2 \mid B^{\#}$. Then

$$(B^{\#})^2 - 4A^{\#}C^{\#} \equiv 0 \pmod 4,$$

since both $(B^{\#})^2$ and $4A^{\#}C^{\#}$ are divisible by 4. Hence $v_2(\mathrm{cont}(\Delta_n^{\#})) \geq 2$, giving $v_2(S(n)) \geq 2e_2 + 2$.

For the upper bound, the same deformation specialization as in §5.3 (at distance $t = 2^{k-1}$) gives $B^{\#} \equiv 2u + 2\mu \pmod 4$ with $2 \nmid u, \mu$. Expanding:

$$(B^{\#})^2 - 4A^{\#}C^{\#} \equiv (2u + 2\mu)^2 - 4u^2 = 4\mu(2u + \mu) \pmod{4 \cdot 2}.$$

Since μ is odd and $2u + \mu$ is odd (even plus odd), $v_2(4\mu(2u + \mu)) = 2$. Combined with the lower bound: $v_2(\mathrm{cont}(\Delta_n^{\#})) = 2$, so $v_2(S(n)) = 2e_2 + 2$, which is even. □

6. The remaining $p = 3$ non-prime-power case

This section handles the $p = 3$ non-prime-power case, namely

$$3 \mid m = n + 2, \qquad m \text{ not a power of } 3.$$

Combining this section with the arguments of §5 recovers the full odd-prime squarefree-kernel formula.

6.1. **The witness family.** For $1 \leq k \leq n - 1$, let ϕ_k denote the symmetric two-point specialization

$$f_k = 1, \qquad f_{N+1-k} = 1, \qquad f_j = 0 \text{ for } j \notin \{k, N+1-k\}. \tag{28}$$

The natural coefficient attached to this specialization is the diagonal B_n-coefficient

$$b_k = [f_k f_{N+1-k}]\, B_n,$$

whose closed formula was proved in Lemma 5.2.

Proposition 6.1. *Let $1 \leq k \leq n - 1$. Under the specialization (28), one has*

$$A_n(\phi_k) = 0, \qquad C_n(\phi_k) = 0, \qquad B_n(\phi_k) = b_k,$$

and hence

$$\Delta_n(\phi_k) = b_k^2.$$

Consequently, if for some such k one has $v_3(b_k) = e_3$, where

$$e_3 := \min_{1 \leq j \leq N} v_3(b_j),$$

then

$$v_3(\Delta_n(\phi_k)) = 2e_3 \qquad \textit{and hence} \qquad v_3(S(n)) \textit{ is even.}$$

Proof. Under (28), every nonzero monomial in A_n has the form $f_i f_{N-i}$, while every nonzero monomial in C_n has the form $f_{i+1} f_{N+1-i}$. Since the support of ϕ_k is $\{k, N+1-k\}$ and $k \neq n, n+1$, neither pattern can hit the same support index twice, so both A_n and C_n vanish.

For B_n, the only surviving monomial is $f_k f_{N+1-k}$, whose coefficient is by definition b_k. Thus

$$\Delta_n(\phi_k) = B_n(\phi_k)^2 - 4A_n(\phi_k)C_n(\phi_k) = b_k^2.$$

If $v_3(b_k) = e_3$, then this specialization exhibits a value of Δ_n with 3-adic valuation exactly $2e_3$. Since every coefficient of Δ_n has valuation at least $2e_3$ (because, exactly as in the proof of Proposition 5.3, every scalar coefficient of A_n, B_n, and C_n has valuation at least e_3), it follows that $v_3(S(n)) = 2e_3$ is even. □

6.2. **Reduction to a binomial maximization problem.** By Lemma 5.2,

$$b_k = 2(-1)^k (N!)^2 \frac{(N+1-k)(N+1-2k)}{\binom{N}{k}}.$$

Now assume $3 \mid m = n+2$, so $N = 2n \equiv 2 \pmod 3$ and $N+1 \equiv 0 \pmod 3$. If $3 \nmid k$, then $N+1-k \not\equiv 0 \pmod 3$ and $N+1-2k \not\equiv 0 \pmod 3$, so both linear factors are 3-adic units. Hence for such k,

$$v_3(b_k) = 2v_3(N!) - v_3\binom{N}{k}. \tag{29}$$

Thus minimizing $v_3(b_k)$ among indices k with $3 \nmid k$ is equivalent to maximizing $v_3\binom{N}{k}$ among those same indices.

Thus the remaining $p = 3$ problem is fundamentally combinatorial: find an off-center lower-half index k with $3 \nmid k$ for which $v_3\binom{N}{k}$ is maximal.

6.3. **Ternary recursions.** Write

$$F_N(k) := v_3\binom{N}{k}.$$

Lemma 6.2. *For integers $N \geq 0$ and $0 \leq k \leq N$,*

$$F_N(k) = \frac{s_3(k) + s_3(N-k) - s_3(N)}{2},$$

where $s_3(t)$ denotes the sum of the base-3 digits of t.

Proof. This is the $p = 3$ specialization of the classical Legendre–Kummer digit-sum formula for binomial coefficients. □

Define also

$$G_A(a) := F_A(a) + v_3(A-a).$$

Lemma 6.3. *For integers $A \geq 0$ and valid values of a, the following hold.*

(i) *If $0 \leq a \leq A$ and $r \in \{0,1,2\}$, then*

$$F_{3A+2}(3a+r) = F_A(a). \tag{30}$$

(ii) *If $0 \leq a \leq A-1$, then*

$$F_{3A}(3a) = F_A(a), \tag{31}$$

$$F_{3A}(3a+1) = G_A(a) + 1, \tag{32}$$

$$F_{3A}(3a+2) = G_A(a) + 1. \tag{33}$$

(iii) *If $0 \leq a \leq A-1$, then*

$$F_{3A+1}(3a) = F_A(a), \tag{34}$$

$$F_{3A+1}(3a+1) = F_A(a), \tag{35}$$

$$F_{3A+1}(3a+2) = G_A(a) + 1. \tag{36}$$

Proof. Part (i) follows immediately from Lemma 6.2: if $N = 3A + 2$ and $k = 3a + r$, then

$$s_3(3a + r) = s_3(a) + r, \qquad s_3(3A + 2 - (3a + r)) = s_3(A - a) + (2 - r),$$

and $s_3(3A + 2) = s_3(A) + 2$.

For part (ii), let $B = A - a > 0$. The identity

$$s_3(B - 1) = s_3(B) - 1 + 2v_3(B)$$

gives

$$s_3(3A - (3a + 1)) = s_3(3(B - 1) + 2) = s_3(B) + 1 + 2v_3(B),$$

and similarly

$$s_3(3A - (3a + 2)) = s_3(3(B - 1) + 1) = s_3(B) + 2v_3(B).$$

Substituting into Lemma 6.2 yields (31)–(33).

Part (iii) is similar. Again setting $B = A - a > 0$, one has

$$3A + 1 - (3a + 2) = 3(B - 1) + 2,$$

so

$$s_3(3A + 1 - (3a + 2)) = s_3(B) + 1 + 2v_3(B).$$

Together with $s_3(3a+2) = s_3(a)+2$ and $s_3(3A+1) = s_3(A)+1$, this gives (36); the identities (34) and (35) are immediate. □

Corollary 6.4. *For integers $A \geq 1$ and valid values of a, one has:*

$$G_{3A}(3a) = G_{3A}(3a + 1) = G_{3A}(3a + 2) = G_A(a) + 1, \tag{37}$$

$$G_{3A+1}(3a + 1) = G_{3A+1}(3a + 2) = G_A(a) + 1, \tag{38}$$

$$G_{3A+2}(3a + 2) = G_A(a) + 1. \tag{39}$$

Moreover,

$$G_{3A+1}(3a) = F_A(a), \qquad G_{3A+2}(3a) = G_{3A+2}(3a + 1) = F_A(a).$$

Proof. This is immediate from Lemma 6.3 and the definition $G_N(k) = F_N(k) + v_3(N - k)$. □

6.4. **Maximizers of the ternary score.**

Lemma 6.5. *For every integer $A \geq 1$, the function $G_A(a)$ $(0 \leq a \leq A - 1)$ attains its maximum at some index satisfying*

$$a < \frac{A}{2}.$$

Proof. We argue by induction on A.

If $A = 1$, then $a = 0$ is the only admissible index.

Assume the claim for all smaller positive integers. If $A = 3B$, then by (37) every maximum of G_A comes from a maximum of G_B lifted to one of the three indices $3b, 3b + 1, 3b + 2$. By induction choose $b < B/2$; then $3b < 3B/2 = A/2$.

If $A = 3B + 1$, then by Corollary 6.4, the values at indices $3b + 1$ and $3b + 2$ are $G_B(b) + 1$, whereas the value at $3b$ is only $F_B(b)$. Thus a maximum of G_A is attained at some $3b + 1$ with b maximizing G_B. By induction $b < B/2$, hence $3b + 1 < (3B + 1)/2 = A/2$.

If $A = 3B + 2$, then similarly the maximal values are attained at indices $3b + 2$ with b maximizing G_B. By induction $b < B/2$, and therefore $3b + 2 < (3B + 2)/2 = A/2$. □

Lemma 6.6. *Let $B \geq 1$ be odd. Then:*

(i) *if $B \neq 2 \cdot 3^t - 1$ for every $t \geq 0$, then $G_B(a)$ attains its maximum at some index satisfying*

$$a < \frac{B - 1}{2};$$

(ii) *if $B = 2 \cdot 3^t - 1$ for some $t \geq 0$, then $G_B(a)$ has a unique maximizer, namely*

$$a = \frac{B - 1}{2}.$$

Proof. We argue by induction on odd B.

For $B = 1$, the only admissible index is $a = 0 = (B-1)/2$, so the statement holds.

Suppose $B > 1$ is odd.

Case $B = 3A + 1$. Then A is even. By Corollary 6.4, the maximal values of G_B occur at indices of the form $3a + 1$ or $3a + 2$, where a maximizes G_A. By Lemma 6.5, choose $a < A/2$. Then

$$3a + 1 < \frac{3A+1}{2} = \frac{B-1}{2}.$$

So part (i) holds in this case. In particular, no number congruent to 1 (mod 3) belongs to the exceptional family.

Case $B = 3A + 2$. Then A is odd. Again by Corollary 6.4, the maximal values of G_B occur precisely at indices $3a + 2$ with a maximizing G_A. If A is not exceptional, the induction hypothesis gives a maximizer $a < (A-1)/2$, hence

$$3a + 2 < \frac{3A+1}{2} = \frac{B-1}{2},$$

so part (i) holds.

If $A = 2 \cdot 3^t - 1$ is exceptional, then by induction $a = (A-1)/2$ is the unique maximizer of G_A. Therefore $3a + 2 = (B-1)/2$ is the unique maximizer of G_B. Since

$$B = 3(2 \cdot 3^t - 1) + 2 = 2 \cdot 3^{t+1} - 1,$$

this is exactly part (ii). □

Lemma 6.7. *Let $A \geq 2$ be even. Then:*

(i) *if $A \neq 2 \cdot 3^t - 2$ for every $t \geq 1$, then $F_A(a)$ attains its maximum at some index satisfying*

$$a < \frac{A}{2};$$

(ii) *if $A = 2 \cdot 3^t - 2$ for some $t \geq 1$, then $F_A(a)$ has a unique maximizer, namely*

$$a = \frac{A}{2}.$$

Proof. We argue by induction on even A.

If $A = 2$, then $F_2(0) = F_2(1) = F_2(2) = 0$, so part (i) holds.

Assume $A > 2$.

Case $A = 3B$. Then B is even. By (32)–(33), the maximal values of F_A occur at indices $3b+1$ or $3b+2$, where b maximizes G_B. By Lemma 6.5, choose $b < B/2$, hence

$$3b + 1 < \frac{3B}{2} = \frac{A}{2}.$$

So part (i) holds.

Case $A = 3B+2$. Then B is even. By (30), the maximal values of F_A are exactly the lifts of the maximal values of F_B. If B is not exceptional, choose $b < B/2$ by induction; then $3b + 1 < (3B + 2)/2 = A/2$, so part (i) holds. If $B = 2 \cdot 3^t - 2$ is exceptional, then $b = B/2$ is the unique maximizer of F_B, but the lifted index

$$3b = \frac{3B}{2} < \frac{3B+2}{2} = \frac{A}{2}$$

is still a strict lower-half maximizer of F_A. Hence part (i) holds in all cases.

Case $A = 3B+1$. Then B is odd. By (34)–(36), the maximal values of F_A occur at indices $3b+2$, where b maximizes G_B. If B is not exceptional, Lemma 6.6 gives $b < (B-1)/2$, so

$$3b + 2 < \frac{3B+1}{2} = \frac{A}{2},$$

and part (i) follows.

If $B = 2 \cdot 3^t - 1$ is exceptional, then by Lemma 6.6 the unique maximizer of G_B is $b = (B-1)/2$, and the corresponding lifted index is

$$3b + 2 = \frac{3B+1}{2} = \frac{A}{2}.$$

Thus $a = A/2$ is the unique maximizer of F_A. Since $A = 3(2 \cdot 3^t - 1) + 1 = 2 \cdot 3^{t+1} - 2$, this is exactly part (ii). □

6.5. Completion of the $p = 3$ non-prime-power case.

Proposition 6.8. *If $3 \mid m = n + 2$ and m is not a power of 3, then $v_3(S(n))$ is even.*

Proof. Set $N = 2n$. Since $3 \mid n + 2$, we may write

$$N = 2n = 3A + 2, \qquad A = \frac{N-2}{3} = \frac{2m-6}{3} = 2\Big(\frac{m}{3}\Big) - 2.$$

Because m is not a power of 3, the integer A is not of the exceptional form $2 \cdot 3^t - 2$. By Lemma 6.7, there exists $a < A/2$ maximizing F_A.

Set

$$k = 3a + 1.$$

Then $1 \le k \le n - 1$, $3 \nmid k$, and by (30),

$$F_N(k) = F_{3A+2}(3a + 1) = F_A(a) = \max_{0 \le b \le A} F_A(b).$$

Hence k maximizes $F_N(j) = v_3\binom{N}{j}$ among all indices $j \not\equiv 0 \pmod 3$, and by (29) it minimizes $v_3(b_j)$ among those same indices.

If $j \equiv 0 \pmod 3$, then $N + 1 - j$ and $N + 1 - 2j$ are both divisible by 3 (because $N + 1 \equiv 0 \pmod 3$), so the closed formula for b_j shows

$$v_3(b_j) \ge 2v_3(N!) + 2 - v_3\binom{N}{j}.$$

Writing $j = 3c$, Lemma 6.3(i) gives $v_3\binom{N}{j} = F_A(c) = F_N(3c + 1)$, so

$$v_3(b_j) \ge v_3(b_{3c+1}) + 2.$$

Thus the global minimum of $v_3(b_j)$ is attained at some index not divisible by 3. Since k minimizes $v_3(b_j)$ among those indices, we conclude that $v_3(b_k) = e_3$.

Now Proposition 6.1 applies: under the symmetric pair specialization ϕ_k one has

$$\Delta_n(\phi_k) = b_k^2 \qquad \text{and} \qquad v_3(\Delta_n(\phi_k)) = 2e_3.$$

Hence $v_3(S(n)) = 2e_3$ is even. □

7. The AI-assisted workflow

We describe the workflow used to complete this project in some detail, as we believe it may serve as a useful model for other computation-heavy research problems.

Starting point. The author's original notes (written several years before the present paper) contained:

- the definition of the test monomial family $M_{n,r}$ and the observation that its coefficient in Δ_n is divisible by $\binom{n+2}{r}$;
- the connection to Lucas' and Kummer's theorems via the binomial gcd $g(m)$;
- the proof that primes dividing n contribute even p-adic valuation;
- the proof for the simplest prime-power case $m = p$ (i.e., $k = 1$);
- various incomplete attempts at the remaining cases.

The main obstacles were: (a) the composite case when $p = 3$, where the natural "end-anchored" witness approach fails; (b) the prime-power case $m = p^k$ for $k \ge 2$, which requires understanding the p-adic structure of the transvectant at depth beyond the leading term; and (c) assembling a complete, gap-free argument from the many partial results.

The workflow. The project was completed in a single extended session using Claude Code (Anthropic), an AI coding assistant accessed via the command line. The workflow had four phases:

Phase 1: Understanding and diagnosis. Claude Code read all of the author's existing notes (three LaTeX files, totaling roughly 10 pages) and produced a detailed inventory of what was proved, what was claimed but not proved, and what was missing. This diagnosis step—which would have taken the author considerable time to reconstruct—was completed in minutes.

Phase 2: Numerical experimentation. Claude Code wrote and executed Python scripts that:

- computed $S(n)$ and $\mathrm{sqf}(S(n))$ for all $n \leq 30$, confirming the prime-to-3 statement of Theorem 1.1;
- verified the closed-form coefficient formula (Lemma 3.2) for all $n \leq 15$ and all valid r;
- investigated the prime-power case by computing p-adic valuations of explicit specializations, discovering that the 4-term specialization with parameters at distance $t = p^{k-1}$ forces odd p-adic valuation;
- confirmed that the cancellation-free B_n coefficient approach works uniformly for all tested primes, and in particular settles the non-prime-power case for $p \geq 5$.

These experiments were essential for guiding the proof strategy: the numerical data revealed the precise mechanism (the first-order p-adic deformation) before the proof was written.

Phase 3: Proof construction and writing. Working from the author's notes and guided by the numerical experiments, Claude Code produced complete proofs of the missing cases, wrote the full paper in LaTeX, and expanded every "similar calculation" and "one verifies" into explicit step-by-step derivations. The author reviewed and corrected the output.

Phase 4: Formal verification. The complete proof was formalized in Lean 4 with Mathlib, using a combination of Aristotle (Harmonic) and Claude Code. The formalization proceeded iteratively: Claude Code prepared Lean files containing theorem statements with `sorry` placeholders and detailed proof sketches, which were submitted to Aristotle for automated proof completion. Aristotle returned either complete proofs or partial progress; in the latter case, Claude Code filled the remaining gaps by launching concurrent agents to work on individual lemmas.

In total, approximately 20 Aristotle submissions were made, of which roughly half returned fully proved files on the first attempt. The hardest lemmas—the Kummer carry analysis for off-centre binomial coefficients and the explicit B_n congruence computation—required multiple rounds. The formal verification also uncovered a sign error in an earlier draft of Proposition 5.3: the valuation shift formula for $v_p(b_{n+2}) - v_p(b_n)$ had the wrong sign in one case, which was corrected using a complete-residue-system argument.

The final formalization is a single self-contained Lean 4 file of approximately 6000 lines, containing zero `sorry` statements and zero `axiom` declarations. It compiles against Mathlib with no errors.

Companion materials. The following companion materials are available at:

`https://github.com/ashvin-swaminathan/quartic-invariant`

- A Jupyter notebook that independently verifies every stated formula and lemma by direct symbolic computation, including the closed-form coefficient formula (Lemma 3.2), the cancellation-free B_n coefficients (Lemma 5.2), and the full main theorem (Theorem 1.1) for all $n \leq 30$.
- A single self-contained Lean 4 file (`Complete.lean`, $\approx$ 6000 lines) that formalizes Theorem 1.1 for all primes $p \geq 2$ (parts (i) and (ii) for $p \geq 5$, plus part (i) for $p = 2$), compiling against Mathlib with zero `sorry` statements and zero `axiom` declarations.

Assessment. The workflow was remarkably effective for this problem. The proof involves extensive algebraic manipulation—tracking signs across four-term sums with factorial coefficients, simplifying products of factorials into binomial coefficients, performing base-p carry analyses—precisely the kind of detail-heavy work that is tedious and error-prone for humans but well-suited to AI assistance. The numerical experimentation phase was particularly valuable: the data unambiguously pointed to the correct proof strategy (the p-adic deformation at distance $t = p^{k-1}$) and ruled out false approaches, before any proof was attempted.

The workflow also has clear limitations. Claude Code occasionally produced algebraic errors that required correction, and some of its initial proof attempts contained logical gaps. The author's mathematical judgment was essential at every stage: for choosing which questions to ask, for evaluating whether proposed arguments were correct, and for understanding the broader mathematical context. The AI did not "discover" the theorem—the pattern was identified by extensive computation years ago—nor did it contribute

any conceptual insight beyond what was already present in the author's notes. Its contribution was primarily organizational and computational: systematically working through the technical details that had prevented the project from being completed.

The author takes full responsibility for all mathematical claims in this paper.

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The AI-assisted workflow described in Section 7 used Claude Code (Anthropic) and Aristotle (Harmonic).

References

[BG14] Manjul Bhargava and Benedict H. Gross. Arithmetic invariant theory. Preprint, 2014.

[Bha04a] Manjul Bhargava. Higher composition laws. I. A new view on Gauss composition, and quadratic generalizations. *Ann. of Math. (2)*, 159(1):217–250, 2004.

[Bha04b] Manjul Bhargava. Higher composition laws. II. On cubic analogues of Gauss composition. *Ann. of Math. (2)*, 159(2):865–886, 2004.

[Bha04c] Manjul Bhargava. Higher composition laws. III. The parametrization of quartic rings. *Ann. of Math. (2)*, 159(3):1329–1360, 2004.

[BST13] Manjul Bhargava, Arul Shankar, and Jacob Tsimerman. On the Davenport–Heilbronn theorems and second order terms. *Invent. Math.*, 193(2):439–499, 2013.

[BV16] Manjul Bhargava and Ila Varma. The mean number of 3-torsion elements in the class groups and ideal groups of quadratic orders. *Proc. Lond. Math. Soc. (3)*, 112(2):235–266, 2016.

[DH71] H. Davenport and H. Heilbronn. On the density of discriminants of cubic fields. II. *Proc. Roy. Soc. London Ser. A*, 322(1551):405–420, 1971.

[Dob25] Edgar Dobriban. Transparent reporting of AI use in scientific writing. *arXiv preprint*, arXiv:2511.18828, 2025.

[Dol12] Igor V. Dolgachev. *Classical algebraic geometry: a modern view*. Cambridge University Press, Cambridge, 2012.

[Ell95] Edwin Bailey Elliott. *An introduction to the algebra of quantics*. Clarendon Press, Oxford, 1895.

[FH91] William Fulton and Joe Harris. *Representation theory: a first course*, volume 129 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1991.

[GY03] J. H. Grace and A. Young. *The algebra of invariants*. Cambridge University Press, Cambridge, 1903.

[KR84] Joseph P. S. Kung and Gian-Carlo Rota. The invariant theory of binary forms. *Bull. Amer. Math. Soc. (N.S.)*, 10(1):27–85, 1984.

[Kum52] Ernst Eduard Kummer. Über die Ergänzungssätze zu den allgemeinen Reciprocitätsgesetzen. *J. Reine Angew. Math.*, 44:93–146, 1852.

[Luc78] Édouard Lucas. Théorie des fonctions numériques simplement périodiques. *Amer. J. Math.*, 1(2):184–196, 1878.

[McM89] J. McMahon. On the expression for the Hessian of a binary quantic in terms of the roots. *Ann. of Math.*, 5(1):17–18, 1889.

[Olv99] Peter J. Olver. *Classical invariant theory*, volume 44 of *London Mathematical Society Student Texts*. Cambridge University Press, Cambridge, 1999.

[Sch32] Arnold Scholz. Über die Beziehung der Klassenzahlen quadratischer Körper zueinander. *J. Reine Angew. Math.*, 166:201–203, 1932.

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